

TYPE 1 · Function with a formula

Template

```
#=>Question 1: <title>
def function_name(param1,param2,param3):
    step1=<formula>
    step2=<formula>
    result=<formula using step1 and step2>
    return result
```

```
param1=float(input("Enter ...: "))
param2=float(input("Enter ...: "))
param3=float(input("Enter ...: "))
answer=function_name(param1,param2,param3)
print()
print("Label:", round(answer,2), "<unit>")
• Exact function name and parameter order from the question.
• float() for decimals, int() for whole numbers.
• return inside the function, print outside it.
```

TYPE 2 · Conditions / categories

Template

```
#=>Question 2: <title>
def function_name(value,flag):
    if value<<exception limit>:
        return <fixed amount>
    elif value<=<limit1>:
        result=<amount1>
    elif value<=<limit2>:
        result=<amount2>
    else:
        result=<amount3>

    if flag=="yes":
        result+=<extra>
    return result
```

```
value=int(input("Enter ...: "))
flag=input("Question? ")
print()
print("Label: $" +str(function_name(value,flag)))
• Exception first with return (e.g. under 5 always free).
• Under 5 = <5 · 5–17 = <=17 · 18–64 = <=64 · 65+ = else
• "$"+str(...) so there's no space after the $.
```

TYPE 3 · Loop with totals & counter

Template

```
#=>Question 3: <title>
total=0
total_cost=0
counter=0

for day in range(1,<N>+1):
    value=float(input("Enter ... day "+str(day)+" : "))
    if value<=<limit>:
        cost=value*<rate1>
    else:
        cost=value*<rate2>
    counter+=1
    total+=value
    total_cost+=cost
```

```
print()
print("Total:", total, "<unit>")
print("Count:", counter)
print("Total cost: $" +str(round(total_cost,2)))
```

Factorial (multiply: start at 1, use *)=

```
def factorial(n):
    result=1
    for number in range(1,n+1):
        result*=number
    return result
```

Analyze numbers (count / add: start at 0, use +=)

```
def analyze_numbers(n):
    even_counter=0
    even_sum=0
    multiple_of_three=0
    for number in range(1,n+1):
        if number%2==0:
            even_counter+=1
            even_sum+=number
        if number%3==0:
            multiple_of_three+=1
    print("Even numbers:", even_counter)
    print("Sum of even numbers:", even_sum)
    print("Multiples of 3:", multiple_of_three)
```

- Totals start at 0 **before** the loop; += inside; print **after**.
- No sum(), max(), min(). Even: number%2==0. Divisible by 3: number%3==0.

Rules for every question

- One .py file, a #=>Question header for each.
- "Define a function": write the def first, then input, then call it.
- Copy the input() prompts and print() labels from the sample run.
- Test the sample input, then another input.
- Only print() input() int() float() range() round().

TYPE 4 · Nested-loop patterns

Base template

```
#=>Question 4: <title>
n=int(input("Enter a positive integer: "))
print()
for row in range(1,n+1):
    for column in range(1,n+1):
        print("",end="")
    print()
```

Which option?

If the sample shows...	Use
Every row same length	range(1,n+1) both loops
Rows get longer	inner range(row)
Rows get shorter	outer range(n,0, -1)
Pushed to the right	spaces loop range(n-row) first
Something alternates	row%2==1 / column%2==1 / (row+column)%2==0
Spaces between (0 0 0)	end=" "
No spaces (***)	end=""

Empty print() lines up with the inner for: it changes the row. If asked for a function, put the loops in a def and call it with the user's number.

A1. Solid square

```
****
****
****

for row in range(1,n+1):
    for column in range(1,n+1):
        print("",end="")
    print()
```

A2. Rows alternate 0/1 (midterm Q4)

```
0 0 0 0
1 1 1 1
0 0 0 0

for row in range(1,n+1):
    for column in range(1,n+1):
        if row%2==1:
            print("0",end=" ")
        else:
            print("1",end=" ")
    print()
```

A3. Columns alternate 0/1

```
0 1 0 1
0 1 0 1

for row in range(1,n+1):
    for column in range(1,n+1):
        if column%2==1:
            print("0",end=" ")
        else:
            print("1",end=" ")
    print()
```

A4. Checkerboard

```
0 1 0 1
1 0 1 0

for row in range(1,n+1):
    for column in range(1,n+1):
        if (row+column)%2==0:
            print("0",end=" ")
        else:
            print("1",end=" ")
    print()
```

A5. Hollow square

```
****
*  *
****

for row in range(1,n+1):
    for column in range(1,n+1):
        if row==1 or row==n or column==1 or column==n:
            print("*",end="")
        else:
            print(" ",end="")
    print()
```

A6. Diagonal of 1s

```
1 0 0
0 1 0
0 0 1

for row in range(1,n+1):
    for column in range(1,n+1):
        if row==column:
            print("1",end=" ")
        else:
            print("0",end=" ")
    print()
```

A7. Multiplication table

```
1 2 3
2 4 6
3 6 9

for row in range(1,n+1):
    for column in range(1,n+1):
        print(row*column,end="\t")
    print()
```

B8. Growing triangle

```
*
**
***

for row in range(1,n+1):
    for column in range(row):
        print("*",end="")
    print()
```

B9. Upside-down triangle

```
***
**
*

for row in range(n,0,-1):
    for column in range(row):
        print("*",end="")
    print()
```

B10. Right-aligned triangle (her Ex3)

```

*
**
***

for row in range(1,n+1):
    for space in range(n-row):
        print(" ",end="")
    for star in range(row):
        print("*",end="")
    print()
```

B11. Right-aligned upside-down

```

***
**
*

for row in range(n,0,-1):
    for space in range(n-row):
        print(" ",end="")
    for star in range(row):
        print("*",end="")
    print()
```

B12. Pyramid

```

*
***
*****

for row in range(1,n+1):
    for space in range(n-row):
        print(" ",end="")
    for star in range(2*row-1):
        print("*",end="")
    print()
```

C13. Counting up 1 / 12 / 123

```
for row in range(1,n+1):
    for column in range(1,row+1):
        print(column,end="")
    print()
```

C14. Row repeats its number 1 / 22 / 333

```
for row in range(1,n+1):
    for column in range(1,row+1):
        print(row,end="")
    print()
```

C15. Decreasing 1234 / 123 / 12 / 1 (her review Ex3)

```
for row in range(n,0,-1):
    for column in range(1,row+1):
        print(column,end="")
    print()
```

C16. Counting down 1 / 21 / 321

```
for row in range(1,n+1):
    for column in range(row,0,-1):
        print(column,end="")
    print()
```

C17. Keeps counting 1 / 2 3 / 4 5 6

```
number=1
for row in range(1,n+1):
    for column in range(row):
        print(number,end=" ")
        number+=1
    print()
```